

EXPERIMENT 1

Linux Programming Environment and basic commands

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AIM

To set up the Linux Programming Environment, install Kali Linux using a Virtual Machine, and learn the basic Linux commands.

A) Setting up of Linux Programming environment

We can do C programming Online or Offline according to the needs. It is required for us to keep working on a Linux Environment.

Online:

Cocalc.ai - A cloud based platform that provides a Linux environment directly in the browser without installation

➤ **Steps to use CoCalc:**

1. Go to cocalc.com
2. Sign up or log in with your email account .
3. Click on create new project and give it a name (e.g. Demo)
4. Open the project and click on New --> Linux Terminal.
5. A Linux terminal opens in the browser where you can write and run C programs.
6. Use commands like nano file.c , cc file.c , ./file to compile and run.

Offline:

There are 4 offline methods to run a Linux Environment:

1. Dual Boot

Installing two operating systems (e.g., windows and Linux) on the same computer. During startup, the user can choose which OS to boot into.

2. Single Boot (Direct Boot)

Only one OS (Linux) is installed on the computer, which boots directly without any option to choose.

3. Bootable Pen drive

Linux OS is installed on a USB pen drive. The computer boots directly from the pen drive without changing the internal OS. Portable and convenient.

4. Virtual Machine (VM)

A virtual Machine allows us to run one OS (guest) inside another OS (host) using software. This is the most used method for beginners.

1. Virtual Machine Manager (Hypervisor)

A **Hypervisor** is a software that lets you run one operating system inside another. It's also called a **Virtual Machine Manager (VMM)**. There are two types:

→ Type-1(Baremetal)

- What is this?
 - It runs **directly on the computer's hardware** – no need for a host operating system. It's fast and mostly used in big servers and data centers.
 - E.g: Microsoft Hyper-V

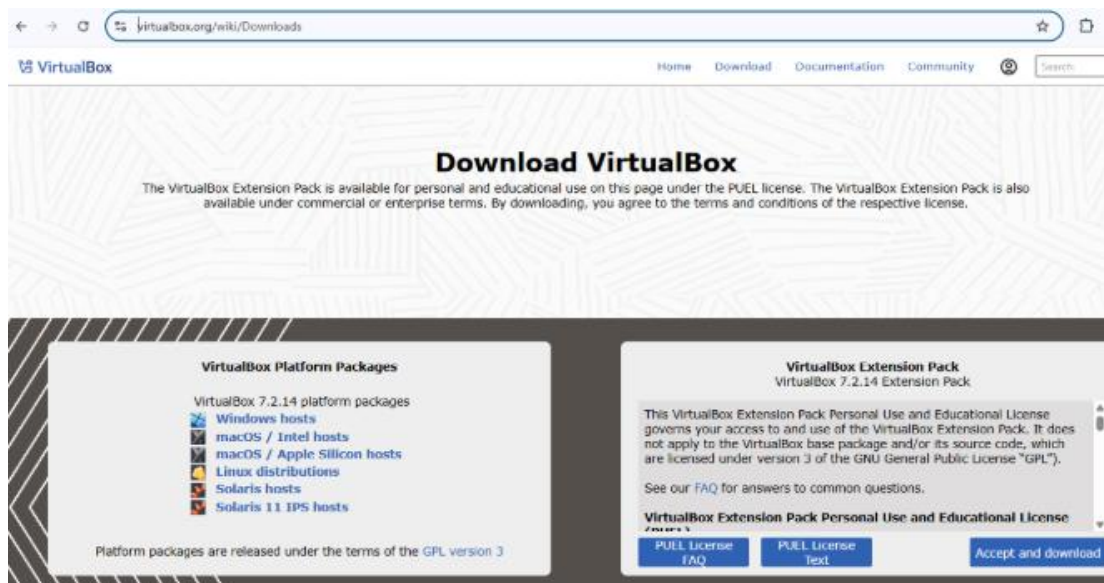
→ Type -2 (Hosted)

- What is this?
 - It runs on top of an existing operating system(like windows or macOS). It's slower than Type-1 but much easier to use—perfect for personal computers and students.
 - E.g: Oracle VirtualBox

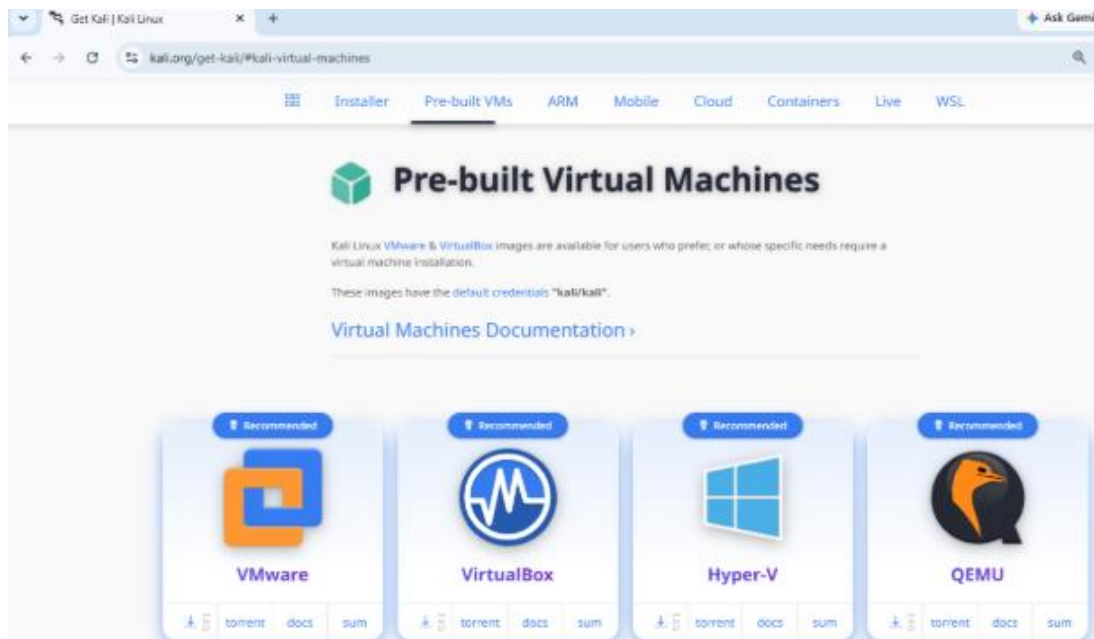
We would be using Oracle Virtual Box.

2. Installation of VMM & VM

Step –1 Download Oracle Virtual Box for “Windows hosts”

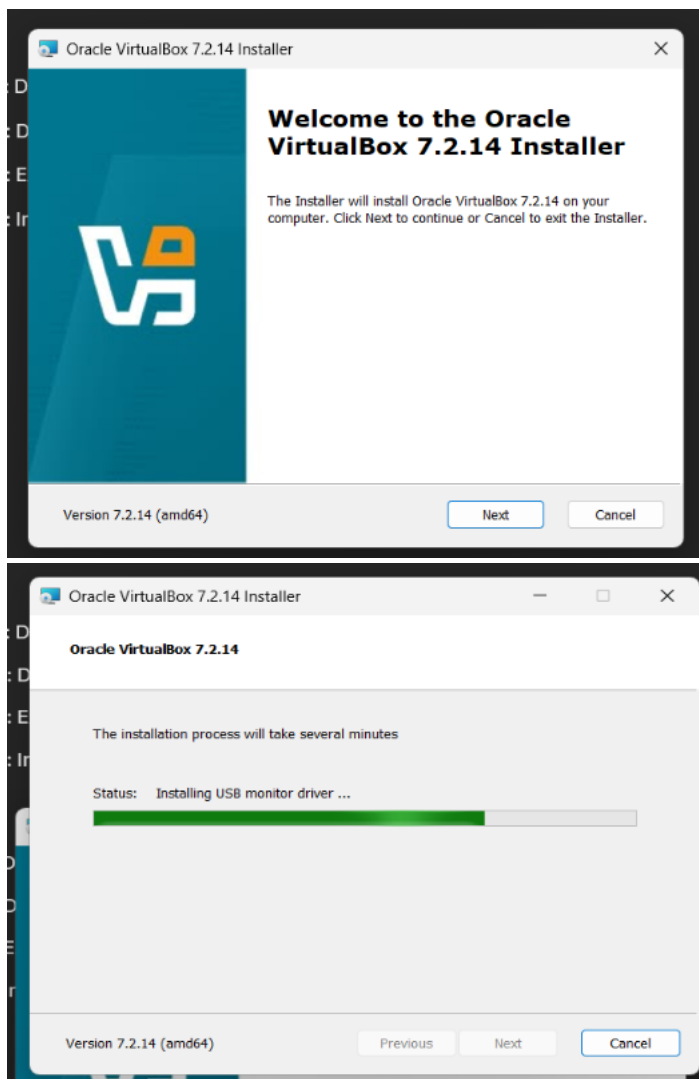


Step-2 Download Pre-built Kali Linux VM for Virtual Box

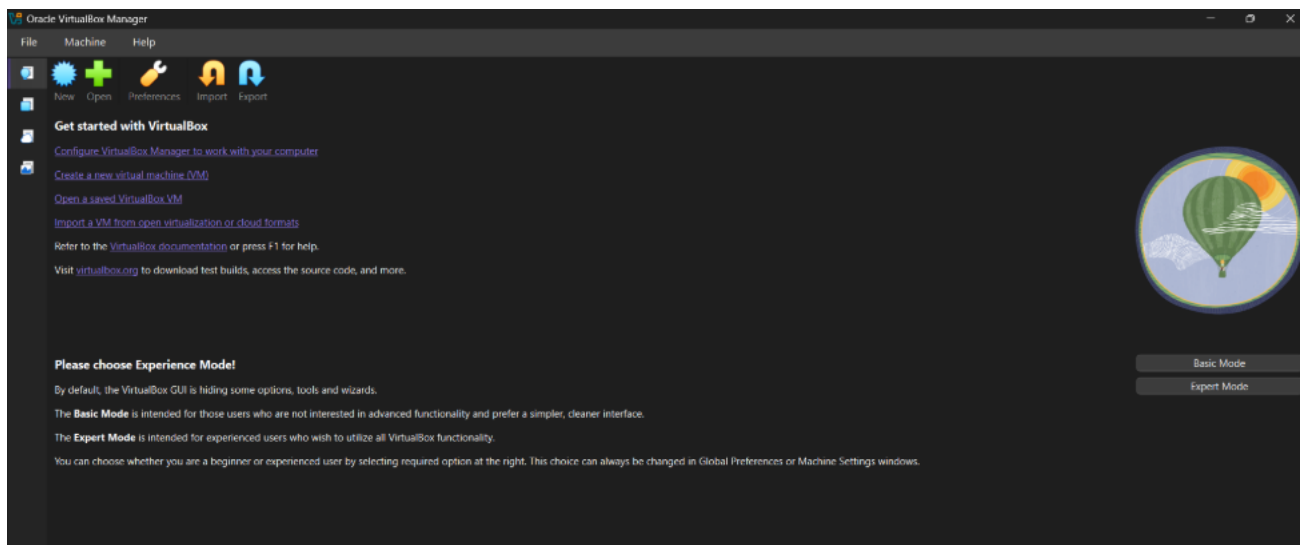


Step-3 Extracted the Kali Linux downloaded file

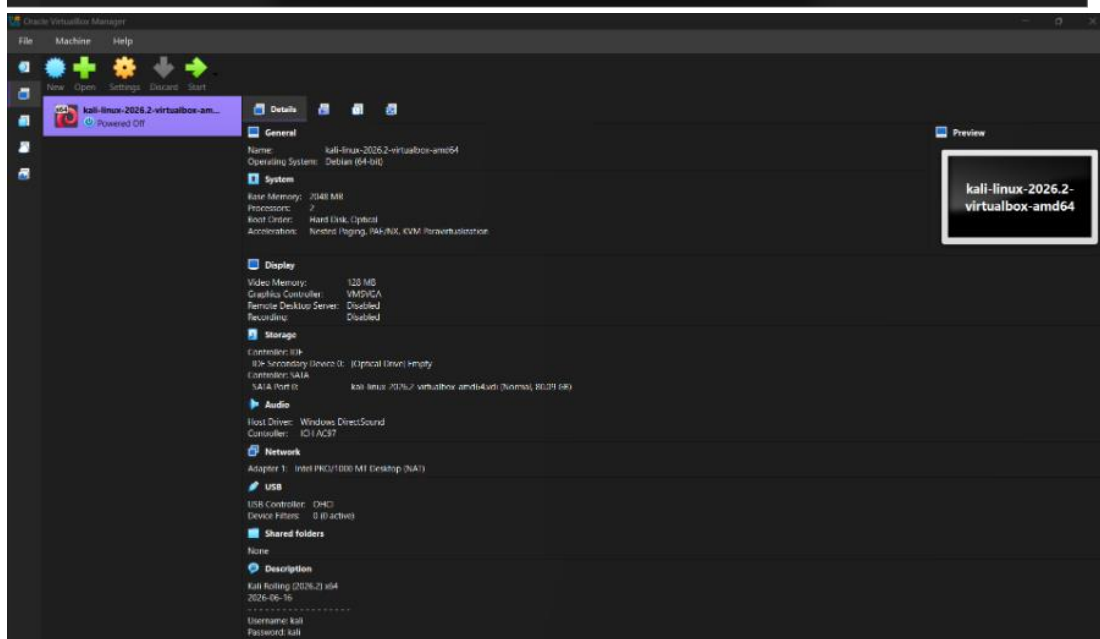
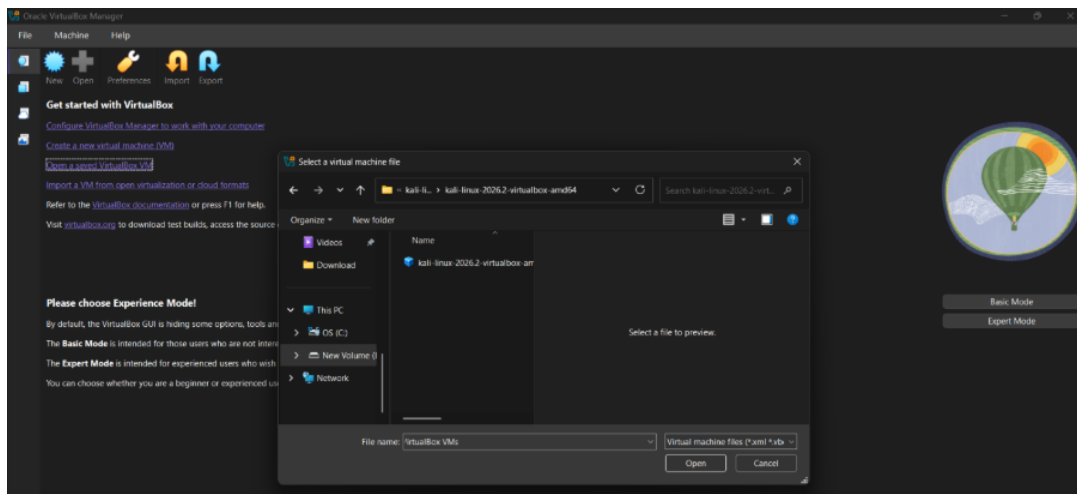
Step-4 Install Virtual Box



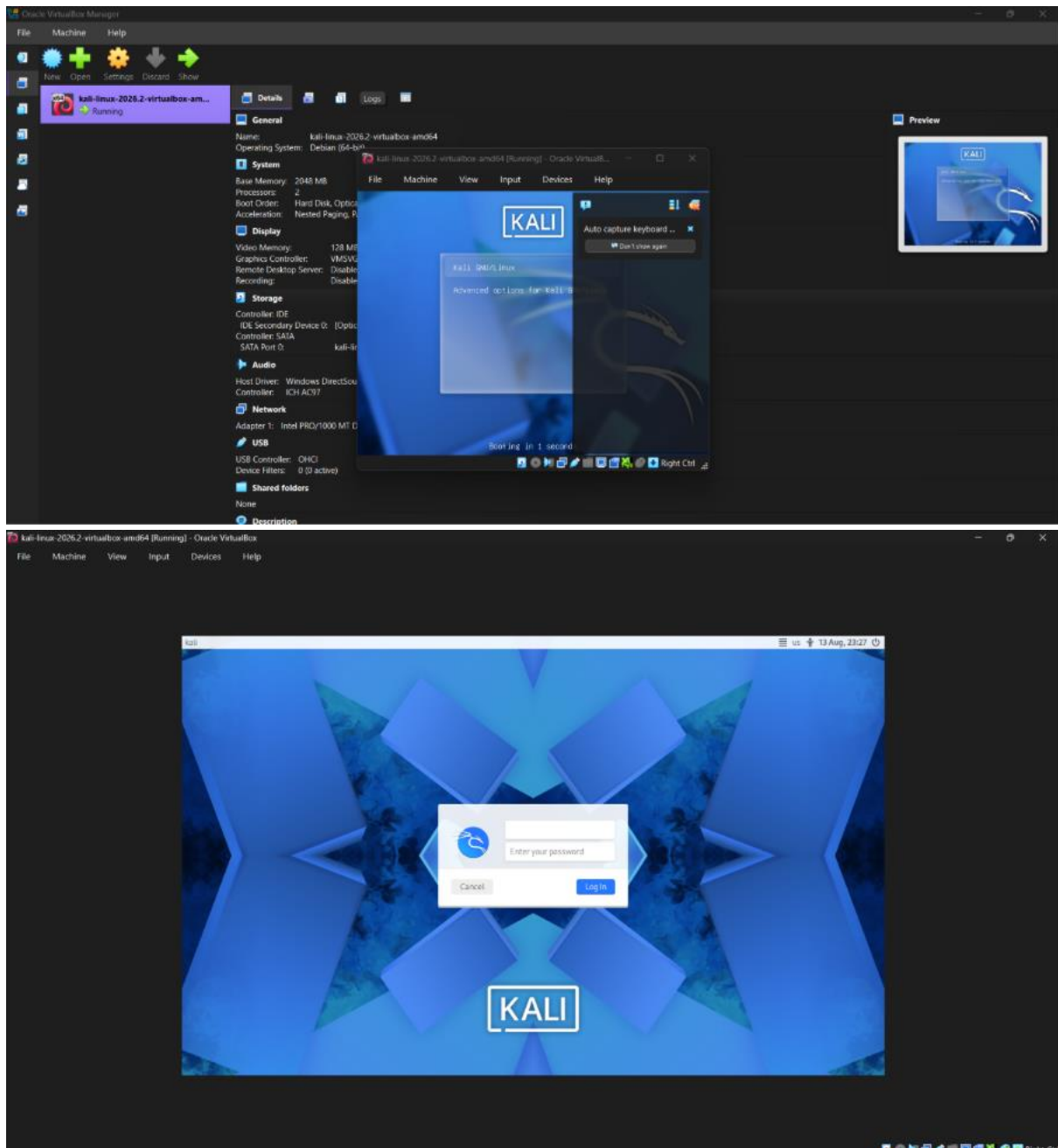
Step-5 Open Virtual Box



Step-6 Open the “saved VM”



Step-7 Start the VM



Thus, the installation of VMM (Hypervisor Type-2 Oracle VirtualBox) & VM Kali Linux) is completed.

Step-8 Open terminal and create the following simple program, compile and execute to understand the working of the environment.

B) Working with Linux Commands

Terminal basics:

1.1 pwd –Print working Directory

- Shows the full path of the current folder you are in.
- Command: pwd
- Output: /Users/puppie/Desktop/experiment1

1.2 mkdir- Make Directory

- Creates a new folder with the given name.
- Command: mkdir testfolder
- Output:(No output – folder created successfully)

1.3 rmdir – Remove Directory

- Removes an empty folder.
- command: rmdir testfolder
- Output: (No output – folder deleted successfully)

1.4 ls /dir- List Files

- Lists all files and folders in the current directory
- Command: ls
- Output: file1.txt mynotes.txt renamed.txt

1.5 cd –Change Directory

- Moves into the specified folder
- Command: cd Desktop
- Output: (prompt changes to show new location)

1.6 cd .. -- Go back one folder

- Moves up the parent directory
- Command: cd ..
- Output: (Returns to the previous folder)

1.7 clear –clear screen

- Clears the terminal screen for a cleaner view.
- Command: clear
- Output:(Blank terminal screen)

1.8 History – Command History

- Displays a list of previously typed commands.
- Command: history
- Output: 1 pwd
2 ls
3 mkdir testfolder
4 cd Desktop
5 clear

1.9 man – Manual

- Shows the manual (help) page of a command
- Command: man ls
- Output : (opens the manual page for ls. Press q to quit)

1.10 whoami – Current User

- Displays the username of the currently logged in user.
- Command: whoami
- Output: puppie

2.Files

2.1 vi- Vi Text editor

- Opens a file in the vi editor to create or edit its contents.
- Command: vi mynotes.txt
- Output: (opens vi editor – press i to type, Esc then :wq to save & quit)

2.2 file –File Type

- Displays the type of the specified file.
- Command: file file1.txt
- Output: file1.txt: ASCII text

2.3 rm – Remove File

- Deletes a file permanently.
- Command: rm file1.txt
- Output: (No output — file deleted successfully)

2.4 cp – Copy File

- Copies a file from source to destination.
- Command: cp file1.txt file2.txt
- Output: (No output — file copied successfully)

2.5 mv – Move / Rename File

- Moves a file to another location or renames it.
- Command: mv file2.txt renamed.txt
- Output: (No output — file renamed successfully)
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2.6 cat – Display Contents

- Displays the entire contents of a file.
- Command: cat file1.txt
- Output:
 - Hello, this is line 1
 - This is line 2
 - This is line 3

2.7 more – Display page by page

- Displays file contents one page at a time.
- Command: more file1.txt
- Output: (Displays file content page by page — press Space for next page, q to quit)

2.8 less – Scrollable Viewer

- Displays file contents with the ability to scroll up and down (better than more).
Command: less file1.txt
- Output: (Displays file with scrolling — press q to quit)

2.9 head – First Lines Displays the first 10 lines of a file.

- Command: head file1.txt
- Output:
 - Hello, this is line 1

This is line 2

This is line 3

2.10 tail – Last Lines

- Displays the last 10 lines of a file.
- **Command:** tail file1.txt
- **Output:** Hello, this is line 1
this is line 2
this is line 3

3.Redirection

3.1 > – Output Redirection (Overwrite)

Sends output to a file. If the file exists, it **overwrites** the content.

Command: echo "Hello" > file.txt

3.2 >> – Output Redirection (Append)

Sends output to a file. If the file exists, it **adds** to the end.

Command: echo "World" >> file.txt

3.3 < – Input Redirection

Takes input **from a file** instead of the keyboard.

Command: wc < file.txt

RESULT

The Linux Programming Environment was successfully set up using Oracle VirtualBox with Kali Linux. All the basic Linux commands (terminal basics, file operations, and redirection) were executed and studied successfully.